

Exercise 4: Functions

Scenario 1: Calculate the age of customers for eligibility checks.

Question: Write a function `CalculateAge` that takes a customer's date of birth as input and returns their age in years.

Program

```
CREATE OR REPLACE FUNCTION CalculateAge(  
    p_dob DATE  
)  
RETURN NUMBER  
IS  
    v_age NUMBER;  
BEGIN  
    v_age := FLOOR(MONTHS_BETWEEN(SYSDATE, p_dob) / 12);  
    RETURN v_age;  
END;
```

Scenario 2: The bank needs to compute the monthly installment for a loan.

Question: Write a function **`CalculateMonthlyInstallment`** that takes the loan amount, interest rate, and loan duration in years as input and returns the monthly installment amount.

Program

```
CREATE OR REPLACE FUNCTION CalculateMonthlyInstallment(  
    p_loanAmount NUMBER,  
    p_interestRate NUMBER,
```

```

    p_years NUMBER
)

RETURN NUMBER

IS

    v_rate NUMBER;

    v_months NUMBER;

    v_emi NUMBER;

BEGIN

    v_rate := p_interestRate / (12 * 100);

    v_months := p_years * 12;

    v_emi := (p_loanAmount * v_rate * POWER((1 + v_rate), v_months))

        / (POWER((1 + v_rate), v_months) - 1);

    RETURN ROUND(v_emi, 2);

END;

```

Scenario 3: Check if a customer has sufficient balance before making a transaction.

Question: Write a function **HasSufficientBalance** that takes an account ID and an amount as input and returns a boolean indicating whether the account has at least the specified amount.

Program

```

CREATE OR REPLACE FUNCTION HasSufficientBalance(

    p_accountID NUMBER,

    p_amount NUMBER

)

```

RETURN BOOLEAN

IS

 v_balance NUMBER;

BEGIN

 SELECT Balance

 INTO v_balance

 FROM Accounts

 WHERE AccountID = p_accountID;

 IF v_balance >= p_amount THEN

 RETURN TRUE;

 ELSE

 RETURN FALSE;

 END IF;

EXCEPTION

 WHEN NO_DATA_FOUND THEN

 RETURN FALSE;

END;